

Inti Shruthi

Hyderabad, Telangana

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LinkedIn

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Portfolio

OBJECTIVE

Electronics and Communication Engineering student with hands-on experience in FPGA Design, Embedded Systems, Artificial Intelligence, Signal Processing, and Software Development through industrial internships and research projects. Passionate about developing intelligent hardware-software solutions and contributing to innovative semiconductor, embedded, and AI-driven applications.

EDUCATION

B.Tech in Electronics and Communication Engineering, Anurag University 2023 – 2027
CGPA: 8.35(till 3-2)

Intermediate (MPC), Sri Chaitanya Junior College 2021 – 2023
Percentage: 91.7%

CBSE Class X, Suprabhat Model High School 2020 – 2021
Percentage: 90%

TECHNICAL SKILLS

- **Programming:** C, Python, MATLAB, Java (Basic)
- **Hardware Design:** VHDL, Verilog HDL (Basic)
- **Embedded Systems:** Xilinx ZC702, Raspberry Pi, Arduino UNO, Microcontrollers, Microprocessors
- **EDA Tools:** Xilinx Vivado Design Suite, Xilinx Vitis, Keil uVision, Arduino IDE
- **Core Domains:** Digital Electronics, FPGA Design, Embedded Systems, Digital Signal Processing (DSP), VLSI Fundamentals

EXPERIENCE

Intern 11 June 2026 – 1 July 2026

Bharat Electronics Limited (BEL), Nacharam, Hyderabad

- Worked with the Xilinx ZC702 FPGA platform and Vivado Design Suite for digital hardware development.
- Designed, simulated, and validated VHDL-based digital modules while gaining practical experience in the FPGA design flow, including synthesis and hardware implementation.

Intern 2026 – Present

Sarvaya Technologies, Hyderabad

- Assisting in the development and validation of technology-driven solutions through feature implementation, testing, and debugging.
- Collaborating with development teams to enhance product functionality and ensure reliable system performance.

Tech Lead Intern May 2025 – July 2025

Swecha Software Pvt. Ltd., Hyderabad

- Coordinated a team of interns in planning, task execution, and technical discussions for AI-based projects.
- Monitored project progress, facilitated collaboration, and ensured timely completion of development milestones.

Familiarization Program Trainee June 2025

Airports Authority of India (AAI), Hyderabad

- Gained practical exposure to airport communication networks, radar systems, Air Traffic Control (ATC) operations, and aviation safety protocols.
- Studied the integration of communication, navigation, and surveillance systems used in modern airport infrastructure.

PROJECTS

Adaptive Active Noise Cancellation & Speech Enhancement Jan 2026 – Mar 2026

- Developed a DSP-based speech enhancement system using NLMS, FxLMS, and MMSE-STSA algorithms.
- Achieved 89% speech enhancement efficiency, validated using SNR, MSE, and spectral analysis.

FPGA-Based CNN Accelerator for Leukemia Cell Classification Jan 2026 – Feb 2026

- Developed a Verilog RTL-based CNN accelerator for FPGA-based leukemia cell classification.
- Achieved 96% accuracy with 20 μ s inference latency through optimized CNN hardware modules.

FPGA-Based Theft Detection System Feb 2025

- Developed an FPGA-based real-time theft detection system using Verilog HDL for intrusion monitoring.
- Verified hardware functionality through simulation, timing analysis, and FPGA testing.

EEG Signal Denoising System Sep 2024 – Feb 2025

- Developed a DSP-based EEG denoising framework to remove motion artifacts and power-line interference.
- Applied digital filtering techniques to improve signal quality while preserving clinically relevant brain activity.

Spiking U-Net for Speech Enhancement 2026

- Developed an SNN-based U-Net model for noise suppression and speech quality enhancement.
- Applied event-driven neural processing for efficient audio enhancement with reduced computation.

Anusandhan – Offline AI Companion Aug 2024 – Apr 2026

- Developed a privacy-focused offline AI assistant using Raspberry Pi and local language models.
- Implemented speech recognition and natural language interaction with a patent-pending system design.

SevaSetu Telangana

- Developed an AI-assisted multilingual platform for accessing verified Telangana government services.
- Integrated OCR-based document processing, eligibility guidance, and AI-driven assistance workflows.

AWARDS & ACHIEVEMENTS

- Won **First Prize** at **TECHSYNAPSE TECHBRIDGE**, a 24-hour National Level Hackathon, for the **FPGA-Based Leukemia Cell Classification** project.
- Won **Third Prize** at the **College Science Expo 2025** for the **Anusandhan – Offline AI Companion Robot**.
- Patent application submitted for the **Anusandhan – Offline AI Companion Robot**.
- Research paper on **Spiking U-Net for Speech Enhancement using Spiking Neural Networks** currently under review.
- Research paper on **Adaptive Active Noise Cancellation & Speech Enhancement using MATLAB** currently under review.

LEADERSHIP & VOLUNTEERING

NodeMCU IoT Workshop Workshop Organizer

- Organized a two-day IoT workshop on NodeMCU and embedded system applications.

Hacktronics 2.0 National-Level Hackathon Volunteer

- Supported event coordination, participant management, and technical sessions.